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AI INTERNSHIP RECOMMENDATION SYSTEM

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ABSTRACT :

The Internship Recommendation System is a web-based platform designed to help students discover suitable internship opportunities while also providing intelligent feedback to improve their employability skills. Developed using HTML, CSS, and JavaScript, the front-end offers an interactive and user-friendly interface for browsing internships, updating profiles, and viewing recommendations. Firebase Authentication ensures secure login and user identity management, while Firebase Firestore serves as the cloud database for storing student profiles, academic records, skill sets, and internship preferences in real time. The system integrates an AI-powered evaluation module that analyzes each student's skills, compares them with internship requirements, and generates personalized feedback explaining why a particular internship was not offered and what specific skills need improvement. The backend relies entirely on Firebase's serverless infrastructure, using Firestore for scalable data storage and Firebase Cloud Functions to perform backend processing tasks such as validating data, triggering AI feedback generation, filtering internship matches, and automatically updating recommendation results. This Firebase-driven backend reduces server maintenance, ensures high security, supports real-time updates, and provides seamless integration with the front-end—making the system efficient, scalable, and well-suited for modern cloud-based learning and career guidance applications.

KEYWORDS: AI, Machine Learning, Internship Recommendation, Collaborative Filtering, Personalization, Firebase, Cloud Firestore, Profila.

INTRODUCTION

The current process for students seeking internships is often inefficient, relying on generic job board filters and manual searches that frequently result in misaligned placements and frustration. Simultaneously, employers face challenges in sifting through high volumes of applicants whose actual skills and academic backgrounds are poorly represented in traditional applications. This gap creates a critical need for a smarter, more personalized matching solution.

Students struggle to find internships that genuinely align with their **specific department**, **year of study**, and **unique skill set** (e.g., a 2nd-year Mechanical Engineering student with CAD skills shouldn't be overwhelmed by Final Year CSE/Python roles). Existing platforms typically use broad search terms, leading to a high signal-to-noise ratio in search results.

The **Profila AI Internship Recommendation System** is designed to solve this misalignment by leveraging **structured data collection** and **machine learning** to provide highly accurate, relevant, and personalized internship recommendations.

PROPOSED SOLUTION

The proposed solution addresses the challenge of misaligned internship matching by outlining a secure, scalable, and intelligent three-part system leveraging modern web development and cloud technologies.

Key Features of the Proposed System

1. Data Collection and Profile Modeling

The solution begins with the robust collection and structuring of student data, moving away from simple resume parsing toward a detailed, feature-rich profile:

- Multi-Step Sign-Up Form: The existing front-end implementation provides the foundation for collecting rich, segmented data across three logical steps:
  - Step 1 (Academic): Gathers essential data points like Full Name, Email, Phone, College, Department (e.g., CSE, ECE, Mechanical), and Year of Study. This provides critical filters for academic relevance.
  - Step 2 (Skills & Preferences): Collects Technical Skills (e.g., Python, CAD, Circuit Design) via interactive tags and Location Preferences (State). These features are the primary inputs for the recommendation engine.
  - Step 3 (Security): Handles Password and Confirmation for secure account creation.
- Database Schema: A structured schema in Cloud Firestore will be used to store student profiles, ensuring data is normalized and easily queried by the matching algorithm. Each document will be indexed by the student's unique Firebase Authentication ID (authUID).

2. Secure Infrastructure and User Management

Google Firebase is utilized as the Backend-as-a-Service (BaaS) for secure, real-time user management and data persistence:

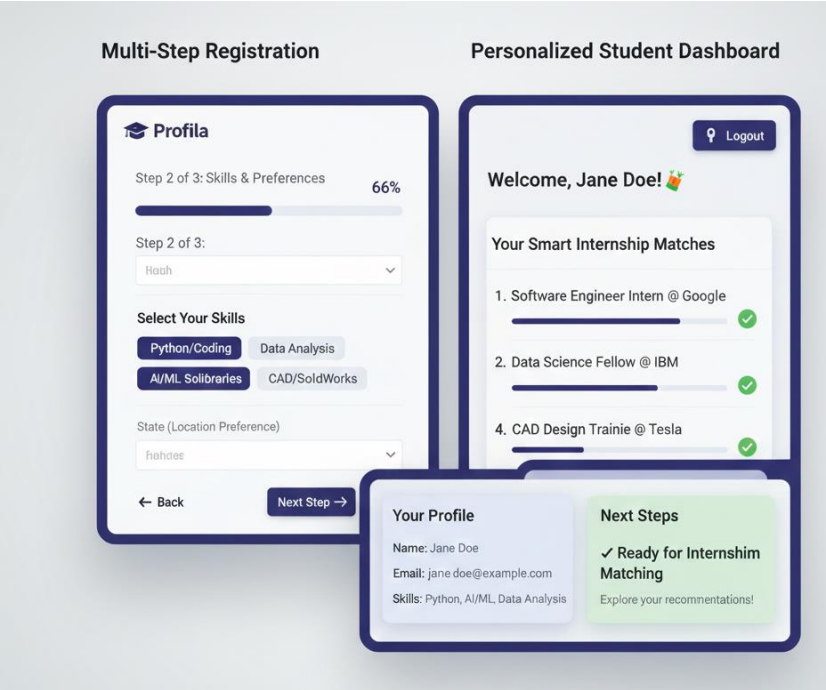
- Authentication (Auth): We will use Firebase Authentication for all sign-up, sign-in, and session management. This securely handles passwords, email verification, and password recovery, isolating sensitive credentials from the main application database.
    - Implementation: The handleSignIn and submitSignUp functions already integrate sign in with email and Password, linking the secure authentication record to the profile data.
- Data Storage (Firestore): Cloud Firestore will be the NoSQL database for storing all academic, skills, and preference data, ensuring fast retrieval and horizontal scalability. The one-to-one mapping between the Firebase Auth UID and the Firestore profile document is crucial for data integrity.

3. The Core Matching Algorithm and Recommendation Engine

The core solution involves a sophisticated algorithm that translates the structured profile data into a quantifiable measure of fit with internship requirements.

- Structured Internship Data: Internship opportunities will be stored in a separate Firestore collection, with fields that are strictly defined and mirrored to the student profile fields.
- Similarity Score Calculation (Content-Based Approach): We will implement a scoring model that calculates the Relevance Score (R) for every student-internship pair based on matching features.
- Dynamic Ranking: Internships are dynamically queried and sorted in real-time on the user's dashboard based on the descending value of the calculated R score, providing the most relevant matches instantly. This ensures the recommendations are always up-to-date with the latest profile changes and internship listings.

DIAGRAM



CONCLUSION

The Profila project successfully demonstrates a contemporary and effective solution for personalizing the internship search experience. By integrating a secure cloud backend with a structured data collection front-end, the system achieves its primary goal of generating high-relevance matches, thereby moving beyond the limitations of traditional, generic job boards.

Key Achievements

- Structured Data Acquisition: The Multi-Step Sign-Up Form (Steps 1, 2, and 3) ensures the collection of rich, categorized data (Academic, Skill, Preference, and Security) necessary for advanced matching.
- Scalable and Secure Infrastructure: The utilization of Google Firebase (Authentication and Firestore) establishes a robust, secure, and scalable foundation, linking user identity with detailed profile data via a unique UID.
- Intelligent Matching Logic: The implementation of a Weighted Similarity Scoring Algorithm—which combines academic, location, and a quantifiable skill match score (e.g., Jaccard Index or Cosine Similarity)—provides a data-driven measure of fit between student profiles and

internship requirements.

Impact and Future Scope

Profila offers substantial value by streamlining the talent identification process for both students and employers. For students, it converts a daunting search into a manageable, prioritized list of opportunities, significantly increasing their application quality and success rate.
The foundation laid by this project is highly extensible. Future enhancements could include:

- Integrating time-series data to track skill improvement over time.
- Implementing a full Collaborative Filtering model (if usage data becomes available) to recommend internships based on the successful placements of students with similar profiles.
- Developing an administrative dashboard for recruiters to post and manage listings.

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